

Multiple Choice

1. **Answer:** D

Options: A) Decreases by a factor of 81.; B) Decreases by a factor of 9.; C) Increases by a factor of 9.; D) Increases by a factor of 81.

Note: Correct option: D - Increases by a factor of 81.

2. **Answer:** C

Options: A) There are no changes in the internal energy of the system.; B) The temperature of the system remains constant during the process.; C) The entropy of the system and its environment remains unchanged.; D) The entropy of the system and its environment must increase.

Note: Correct option: C - The entropy of the system and its environment remains unchanged.

3. **Answer:** D

Options: A) 2; B) 4; C) 6; D) 10

Note: Correct option: D - 10

4. **Answer:** D

Options: A) $mc/2$; B) $mc/(2^{1/2})$; C) mc ; D) $(3^{1/2})mc$

Note: Correct option: D - $(3^{1/2})mc$

5. **Answer:** D

Options: A) $0.1 \text{ GeV}/c^2$; B) $0.2 \text{ GeV}/c^2$; C) $0.5 \text{ GeV}/c^2$; D) $1.0 \text{ GeV}/c^2$

Note: Correct option: D - $1.0 \text{ GeV}/c^2$

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'interactions with the ionic lattice'.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '5'.

9. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: 'Electron'.
10. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** Constant volume. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.
12. **Answer:** $n = 3, l = 1$. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key